Socioeconomic impacts of Terrestrial Protected Areas

Evidence from Large-Scale National Surveys in Madagascar

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Context

To what extent do protected areas (PA) impact household living standards?

▷ Theory of change

- **Economic Opportunities**
  - Development of local tourism (Naidoo et al. 2019),
  - Creation of conservation-related jobs (Balietti et al. 2024)
  - Compensation measures (Poudyal et al. 2018)
  - Improved ecosystem services (Xu et al. 2023)
- **Restrictions and Costs** Restricted access to natural resources - gathering, hunting, fishing, medicinal plants (Kandel et al. 2022)
- **Heterogeneous results** Effects vary across evaluation methods, institutional contexts, and geographic locations (Kandel et al. 2022).

How would household living conditions have evolved if PA had not been established?

Research Design

- **Q1 :** Do PA worsen the living conditions of local communities?
- **Q2 :** Do PA increase inequality among local communities?
- **Q3 :** Do the impacts vary by type of PA?

Hypothesis

- **H1 :** PA in Madagascar may reduce well-being by limiting access to natural resources.
- **H2 :** PA may increase inequality through uneven access to benefits.
- **H3 :** Impacts are heterogeneous depending on governance and context.

Madagascar is a relevant study case

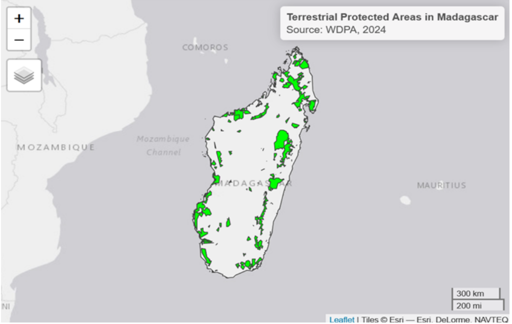
▷ Evolution of protected areas

Surface of PA and Size of the adjacent population exposed to them nearly tripled between 2008 and 2021

- 2008: 3,6% of the territory — 9% population lived $<$ 10 km of PAs
- 2021: 10,8% du territoire — 28% population lived $>$ 10 km of PAs

A vulnerable socio-economic context

- The world's poorest country based on SDG Indicator 1.1 (Conceição 2024)
- High dependence on natural resources (Llopis et al. 2023)
- Weak state capacity (Hanson and Sigman 2021)



Source : Authors

Empirical methodology

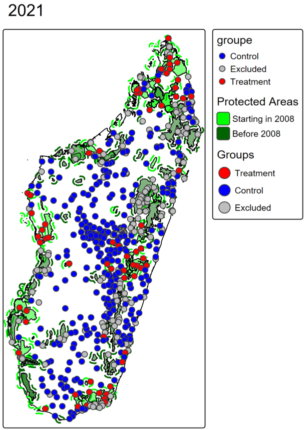
The data at the rural level

▷ "Summary statistics

Role in the analysis	Variable names	Data source
Treatment variable	Protected Areas	*World Database on Protected Areas* (2025)
Outcome variable	Indice niveau de vie Zscore of the wealth index	*DHS/MIS/MICS* (1997, 2008, 2011, 2016, 2018, 2021)
Matching variables	Forest cover rates in 2000 Population density in 2000 Accessibility (2000) Slope and elevation	*Global Forest Change* [@Hansen2013] *WorldPop* [@WorldPop2018] *JRC* [@Uchida2011] NASA-SRTM [@NASAJPL2020]
Control variables	Sex and age of the head of household Drought index (SPEI) [@VicenteSerrano2010]	*DHS/MIS/MICS* (1997, 2008, 2011, 2016, 2018, 2021)

Counterfactual analysis

- **Treatment**: Establishment of PA between 2008 and 2021
- **Treatment group**: Rural Households located in clusters < 10 km of PA created between 2008 and 2021
- **Control group**: Rural Households located in clusters > 10 km of PA created between 2008 and 2021
- **Excluded group**: Households located in clusters within < 10 km of PA created before 2008 and in urban areas



Source: Authors based on DHS, WDPA data

Estimating the impact of PA on living condition

- **Genetic matching**: To compare observable characteristics between the treatment group and the control group (Diamond and Sekhon 2013) ▷ Matching
- **Difference-in-difference (DID)**

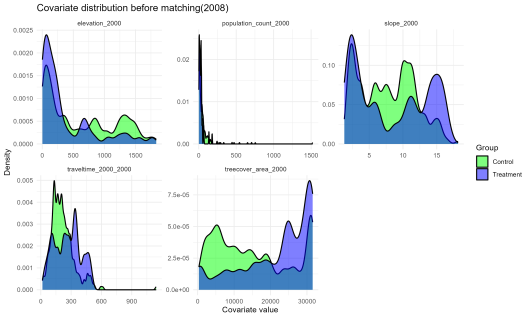
$$Y_{ict} = \alpha + \delta \cdot (\text{Treatment}_c \times \text{Post}_t) + X'_{ict}\theta + \mu_c + \lambda_t + \varepsilon_{ict}$$

- Y_{ict}: wealth index (hh i, cluster c, year t)
- δ: DID effect (PA $\rightarrow$ welfare)
- Treat$_c$ $\times$ Post$_t$: DID term
- μ_c, λ_t: time FE (common shocks)
- X_{ict}: controls (age/sex of the head of household, SPEI, accessibility, population density, forest cover rate, slope, altitude)

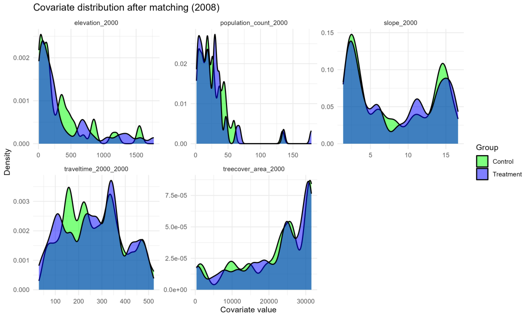
Results

Matching

Before 2008

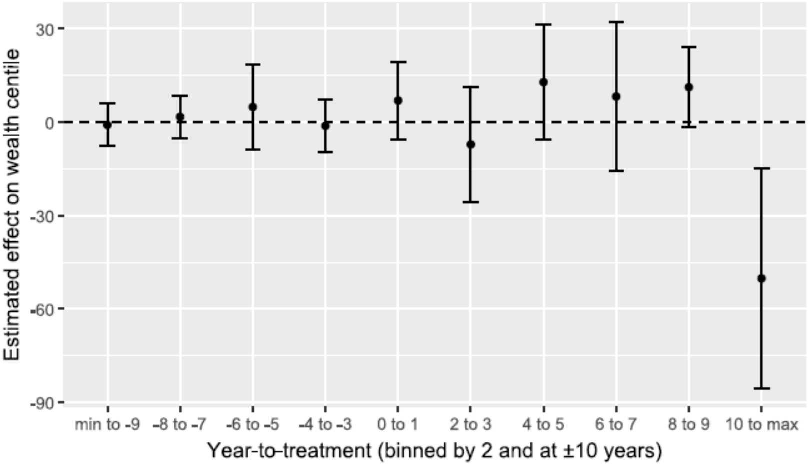


After matching 2008



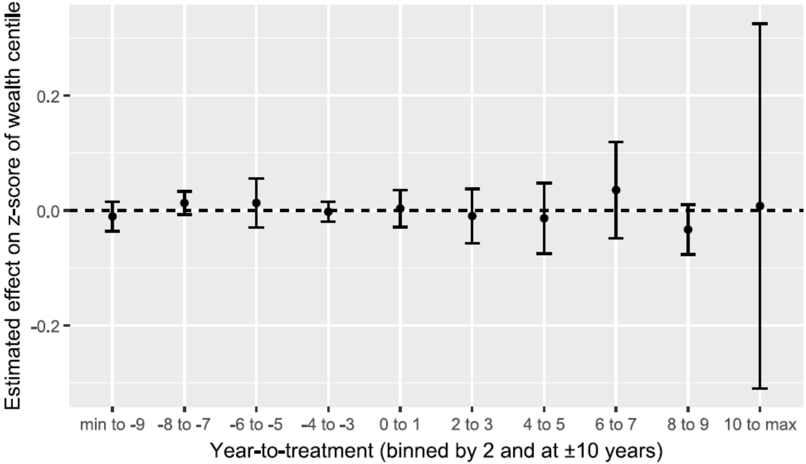
PA impact on the household wealth index

▷ Impact on well-being: Event study with 2 year bins



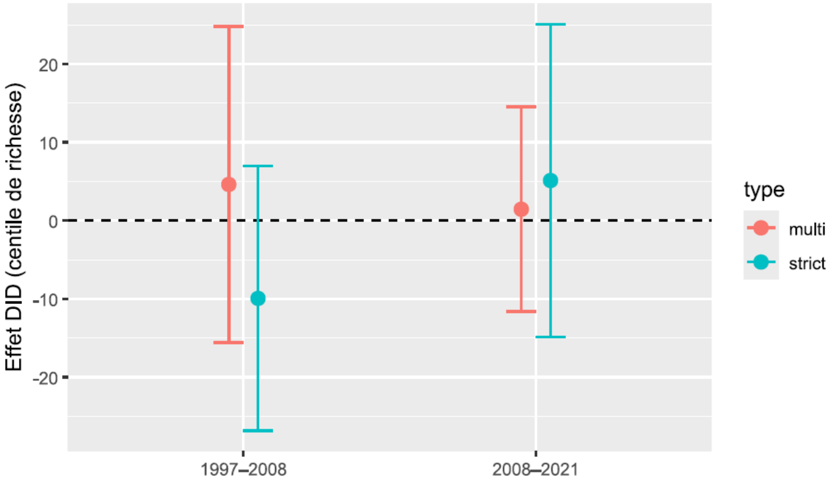
PA impact on the household wealth index inequalities

▷ ”Impact on inequalities: Event-study with 2-year bins



Heterogeneity: Which PA Harm Most?

▷ "Heterogeneity of impact on well-being



Robustness Test

- **Distance threshold test of 5 km and 15km**: Proximity to protected areas (PA) has no significant effect on wealth
- **Pseudo-panel analysis**: the results confirm the robustness of the conclusions in that environmental and spatial conditions have a stronger influence on household living standards than sociodemographic characteristics
- **Heterogeneity analysis**: The effects are not robust to the presence of unobserved bias
- **Other outcome variables (Child size/weight and infant mortality)**: No significant effect is found in the estimation

Conclusions

- PA do not lower the standard of living, but costs accumulate over time
- No significant average effect on the standards of living - losses associated with restrictions appear to be offset by adaptation mechanisms and local opportunities
- No significant effect on inequality - the main determinants remain climate shocks and sociodemographic characteristics
- No differentiated effect based on governance type - formal categories alone do not explain socioeconomic outcomes

Policy Implications: - Integration of systematic socioeconomic monitoring into protected area governance frameworks - Strengthening of transparent compensation mechanisms - Design of inclusive benefit-sharing mechanisms that address equity concerns

Thank you for your attention

Questions & Discussion

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Appendix

Theory of Change

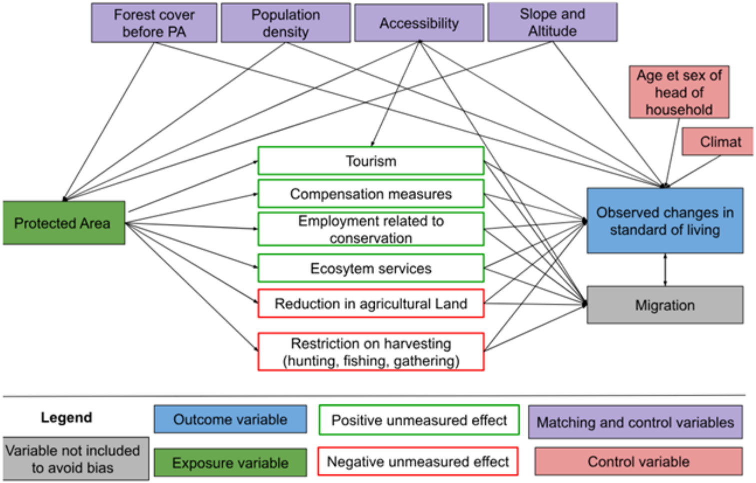
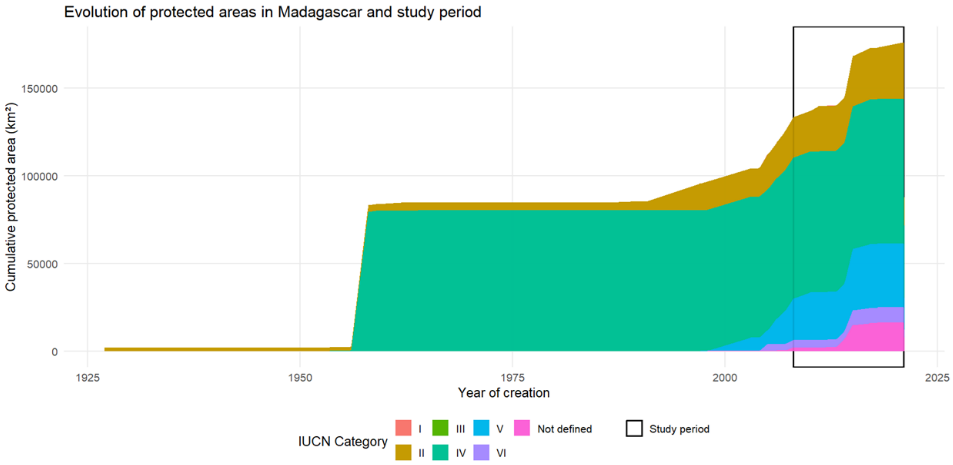


Figure 1: Logic diagram of the theory of change

Evolution of protected areas



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Summary statistics: Distribution of the rural wealth index

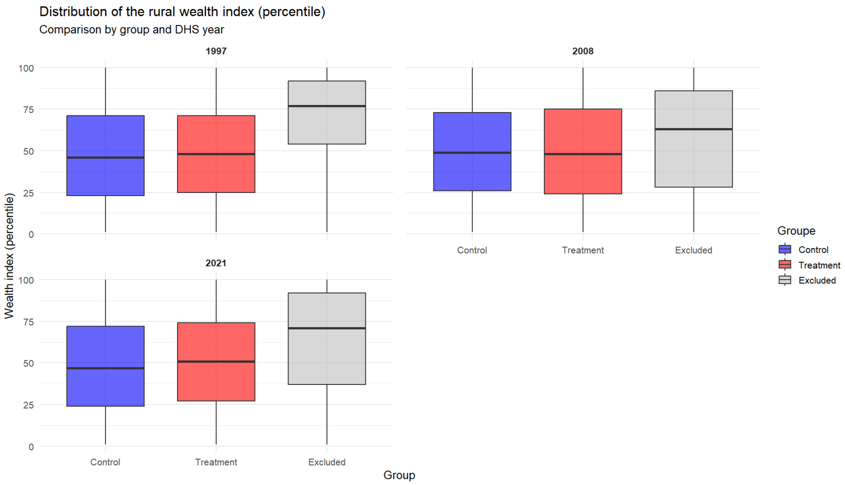

Figure 2: Distribution of the rural wealth index (percentile)

Summary statistics: Distribution of the Z-score of the rural wealth index

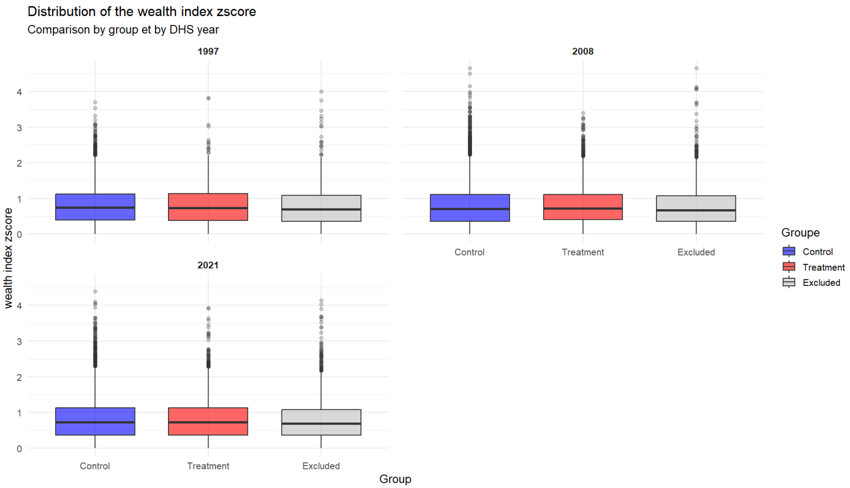

Figure 3: Distribution of the Z-score of the rural wealth index (percentile)

Descriptive statistics

▷ Descriptive statistics

Table 1: Descriptive statistics for protected areas created before 2008 and from 2021

periode	n	mean	sd	median	min	aire_min	max	aire_max
From 2008	84.00	574.17	1,017.13	205.27	0.28	Nosy Antsoha	5,688.62	Complexe des AP Ambohimirahavavy Marivorahona
Before 2008	60.00	2,087.61	10,070.25	247.69	0.05	Parc de Tsarasaotra	78,139.00	Ambatovaky

Matching

Table 2: Standardized Mean Difference of the Year 2008 and 2021

Covariates balance before and after matching

	2008		2021	
Variable	Before matching	After matching	Before matching	After matching
Elevation (m)	0.568	0.056	0.433	0.092
Population density in 2000 (km^2)	0.976	0.050	1.172	0.086
Slope (%)	0.105	0.018	0.150	0.012
Accessibility in 2000 (min)	0.365	0.018	0.379	0.047
Forest cover rates in 2000 (%)	0.729	0.007	0.925	0.101

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Impact on well-being: Event-study with 2-year bins

Table 3: Static treatment effect estimates on well-being

	did2s (static)
treat_on = 1	4.519
	(4.234)
Num.Obs.	4246
R2	0.006
R2 Adj.	0.006
AIC	41025.8
BIC	41032.2
RMSE	30.32
Std.Errors	Corrected Clustered (cluster_uid)

Impact on inequalities: Event-study with 2-year bins

	did2s (static)
treat_on = 1	-0.007
	(0.019)
Num.Obs.	4246
R2	-0.000
R2 Adj.	-0.000
AIC	7251.5
BIC	7257.9
RMSE	0.57
Std.Errors	Corrected Clustered (cluster_uid)

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